IN1006 Systems Architecture

Week 1 Tutorial questions: Introduction to Computer Systems

Exercises

1. Complete Session 1 self-check online tutorial questions (Moodle)

2. What is the definition of Systems Architecture? Why do all Computing students need to understand this topic?

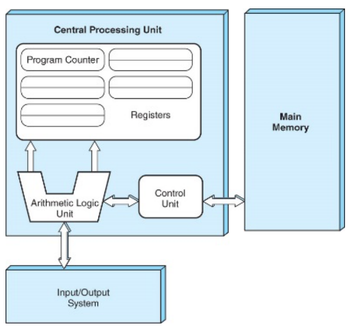
   It is the fundamental organisations of a computer system, embodied in its components, their relations to each other component and the environment, and the principles governing its design and evolution.

   It enables an understanding of the fundamental capabilities of a computer system, how these capabilities are realised by the components and how a computer system may be altered and extended and what effects such actions may have on its own capabilities.

3. Explain layers of abstraction in computer systems?

Computer systems can be decomposed into layers, the process of abstraction allows layers to hide lower level information from higher levels in a way that allows the higher levels to be useful without being cluttered by unnecessary detail.

4. Draw and explain Von Neumann model with its five components and data paths.



All modern stored program computers are based on the von Neumann model. It consists of five components

1. Control unit
2. ALU
3. Registers
4. Main Memory System
5. I/O system

These computers have the capacity to carry out sequential instruction processing